Data-free Neural Representation Compression with Riemannian Neural Dynamics

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Why we need data-free compression

- We target accurate compression when no real data are available.
- We face limits of Euclidean weight linearity and no calibration signals.
- We aim to cut latency and energy while preserving accuracy.

We propose RieM: weights as Riemannian neuron interactions

- We represent each neuron with a D_q-dimensional state and define weights as $g(\mathbf{q}_{\text{in}}, \mathbf{q}_{\text{out}})$.
- We aggregate D_μ metric channels for higher expressivity per parameter.
- We compress by learning neuron states plus a small set of metric parameters, without using real data.

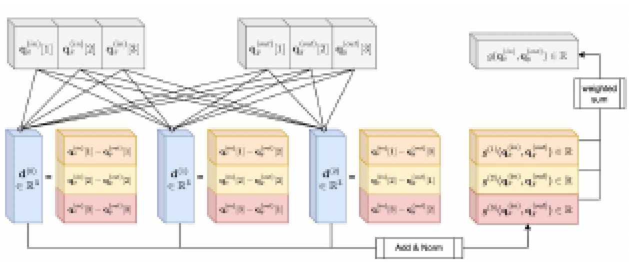

Figure: Neural Riemannian Metric: we express each weight as an interaction $g(\mathbf{q}_{\text{in}}, \mathbf{q}_{\text{out}})$ over neuron states.

How we compress without data

- We optimize $\mathbf{q}_{\text{in}}, \mathbf{q}_{\text{out}}$, and metric parameters to reconstruct pretrained W by minimizing $\|W - g(\mathbf{q}_{\text{in}}, \mathbf{q}_{\text{out}})\|_F$.
- We use a displacement-based, vectorizable metric (Option 2) with nonlinear aggregation for stronger capacity.
- We add SCC to jointly fit correlated matrices and dyMerge to merge redundant output neurons at inference.
- We quantize neuron states/metric outputs to target bit-widths and tune D_q/D_μ to meet size/FLOPs budgets.

Experimental setup

- Backbones: ResNets (18/50), DenseNet-121, Swin-T/S, DETR-ResNet50.
- Tasks: ImageNet-1k, COCO; Metrics: Top-1, AP/AP50/AP75, APS/APM/APL.
- Baselines: Data-free quantization/pruning and ViT quantizers (e.g., UDFC, SQuant, PSAQ-ViT V2).

Main result: we outperform data-free baselines on ImageNet CNNs

- We match storage/compute and achieve higher Top-1 (e.g., ResNet-50 at ∼12.28 MB: we show 73.26% vs 72.09% UDFC, 70.80% SQuant).
- We improve accuracy without increasing FLOPs/BOPs.

Table: ImageNet CNNs: Top-1 accuracy under matched sizes and bit-widths.

Model	Method	Data-free	Size (MB)	W/A-bit	Top-1 (%)
ResNet-18	Original	×	46.83	32/32	71.47
	UDFC (Bai et al., 2023)	✓	8.36	6/6	72.70
	RieM (ours)	✓	8.36	8/16	71.80
ResNet-50	Original	×	102.53	32/32	77.72
	SQuant (Guo et al., 2022)	✓	12.28	4/4	70.80
	UDFC (Bai et al., 2023)	✓	12.28	4/4	72.09
	LP-norm (Pei & Wang, 2023)	✓	12.28	8/16	72.96
	RieM (ours)	✓	12.28	8/16	73.26
DenseNet-121	Original	×	32.34	32/32	74.36
	UDFC (Bai et al., 2023)	✓	6.00	4/32	70.15
	LP-norm (Pei & Wang, 2023)	✓	6.00	8/16	71.66
	RieM (ours)	✓	6.00	8/16	73.15

Limitations

- We require per-layer fitting time; we are developing faster solvers.
- Our metric design space is large; we plan to explore structured and skew-symmetric forms.
- We tune merge thresholds to balance speed and accuracy.

We generalize to Vision Transformers

- At matched size/BOPs, we meet or surpass PSAQ-ViT V2 (e.g., Swin-T 4/8: 76.30%).
- At 8/8 bits, we raise the ceiling (e.g., Swin-S 8/8: 82.96%).

Table: Vision Transformers on ImageNet at matched sizes and BOPs.

Model	Method	Data-free	W/A-bit	Size (MB)	BOPs (G)	Top-1 (%)
Swin-T	Original	×	32/32	116	4608	81.35
	Standard	×	4/8	14.5	144	70.16
	Standard V2	×	4/8	14.5	144	75.51
	PSAQ-ViT	✓	4/8	14.5	144	71.79
	PSAQ-ViT V2	✓	4/8	14.5	144	76.28
	RieM (ours)	✓	4/8	14.5	144	76.30
Swin-S	Original	×	32/32	200	8909	83.20
	Standard	×	4/8	25	278	73.33
	Standard V2	×	4/8	25	278	78.22
	PSAQ-ViT	✓	4/8	25	278	75.14
	PSAQ-ViT V2	✓	4/8	25	278	78.86
	RieM (ours)	✓	4/8	25	278	79.84
Swin-T (higher bits)	PSAQ-ViT V2	✓	8/8	29	288	80.21
	RieM (ours)	✓	8/8	29	288	80.85

Takeaways

- We reframe weights as Riemannian interactions, enabling data-free compression.
- We deliver a practical pipeline (RieM + SCC + dyMerge) with state-of-the-art data-free accuracy across CNNs, ViTs, and DETR.
- We provide strong accuracy at matched budgets with fewer parameters.